ANALYSIS
OF ANALOGUE SYSTEMS

Chapter 03

Recap : System Modelling

SUMMARY – SYSTEM MODELLING

1st Order System

$$x(t) = m\frac{\partial yt)}{\partial t} + ny(t) \qquad H(s) = \frac{a}{s+b}$$

2nd Order System

$$x(t) = p\frac{\partial^2 y(t)}{\partial t^2} + q\frac{\partial y(t)}{\partial t} + ry(t) \qquad H(s) = \frac{a}{s^2+bs+c}$$

3rd Order System

$$x(t) = u\frac{\partial^3 y(t)}{\partial t^3} + v\frac{\partial^2 y(t)}{\partial t^2} + w\frac{\partial y(t)}{\partial t} + zy(t) \qquad H(s) = \frac{g}{s^3+ds^2+es+f}$$

Recap : Lab04 Briefing

SYSTEM MODELLING

1st Order System

$$x(t) = m\frac{\partial yt)}{\partial t} + ny(t) \qquad H(s) = \frac{a}{s+b}$$

2nd Order System

$$x(t) = p\frac{\partial^2 y(t)}{\partial t^2} + q\frac{\partial y(t)}{\partial t} + ry(t) \qquad H(s) = \frac{a}{s^2+bs+c}$$

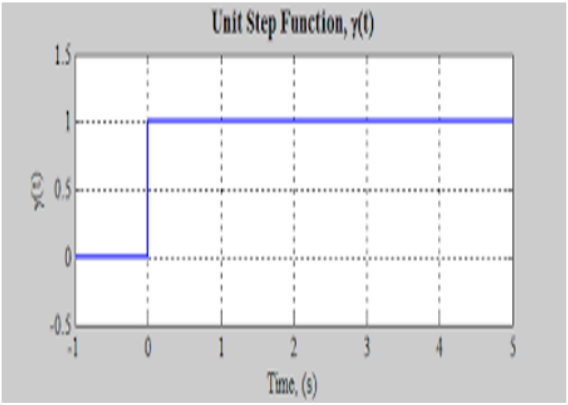
3rd Order System

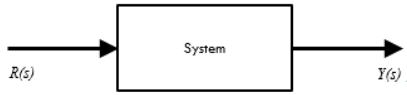
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Position Control of DC Motor

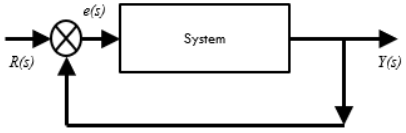
Recap : Lab04 Briefing

STEP RESPONSE





Transfer Function : $G(s) = \frac{Y(s)}{R(s)}$

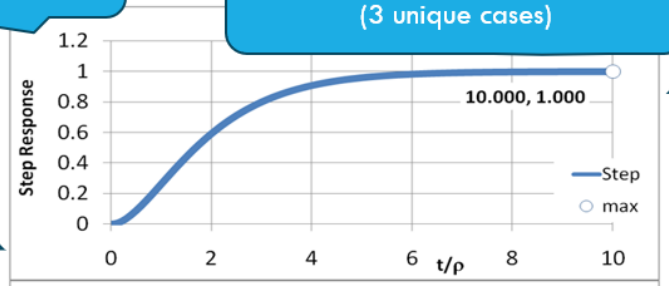


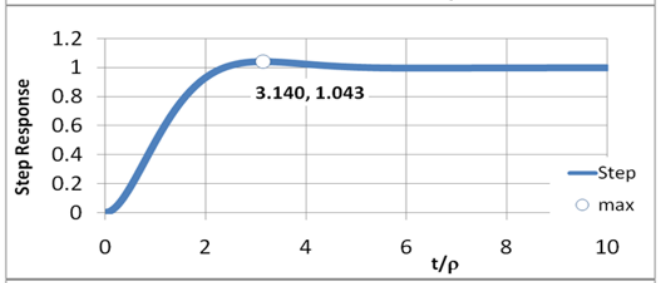
Transfer Function : $T(s) = \frac{G(s)}{1 + G(s)}$

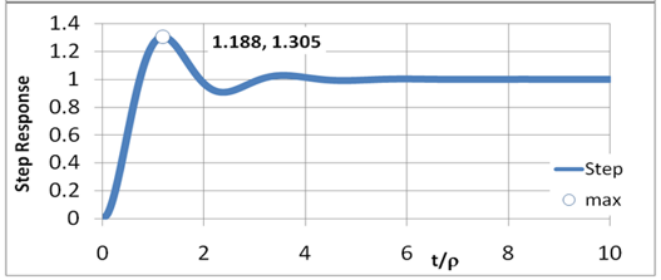
1st Order System
(1 unique case)

2nd Order System
(3 unique cases)

3rd Order System
(5 unique cases)







For Knowledge

STEP RESPONSE OF A 1ST ORDER SYSTEM

1st Order System

$x(t) = m\frac{\partial yt)}{\partial t} + ny(t)$ $\qquad H(s) = \frac{a}{s+b}$

$$v_i(t) = RC\frac{\partial v_o(t)}{\partial t} + v_o(t) \quad \Leftrightarrow \frac{1}{RC}v_i(t) = \frac{\partial v_o(t)}{\partial t} + \frac{1}{RC}v_o(t)$$

If we replace RC and k/b by τ, and set the input ($e_{in}(t)$ and $x_{in}(t)$) to a unit step (γ(t)), and call the output "y" the equations become identical

$$\dot{y}(t) + \frac{1}{\tau}y(t) = \frac{1}{\tau}\gamma(t)$$

We now proceed with the solution:

Homogeneous Solution

We assume a homogeneous solution of the form

$$y_h(t) = Ae^{st}$$

and put this into the differential equation (with the input set to 0) to get

$$\dot{y}_h(t) + \frac{1}{\tau}y_h(t) = 0$$

$$sAe^{st} + \frac{1}{\tau}Ae^{st} = 0$$

$$s = -\frac{1}{\tau}$$

$$y_h(t) = Ae^{-t/\tau}$$

where the coefficient A is yet to be determined.

For Knowledge

STEP RESPONSE OF A 1ST ORDER SYSTEM

1st Order System

$$x(t) = m\frac{\partial yt)}{\partial t} + ny(t) \qquad H(s) = \frac{a}{s+b}$$

Particular Solution

We assume that the particular solution is a constant (since the input is constant for t>0).

$$y_p(t) = K$$

$$\dot{y}_p(t) + \frac{1}{\tau}y_p(t) = \frac{1}{\tau}$$

$$y_p(t) = 1$$

Complete Response

The complete response is simply the sum of the homogeneous and particular responses.

$$y_c(t) = y_h(t) + y_p(t)$$
$$= Ae^{-t/\tau} + 1$$

We find "A" from initial conditions at $t=0^+$. For the electrical circuit $e_{out}(0^+)=0$ (voltage across a capacitor cannot change instantaneously), and $x_{out}(0^+)=0$ (dashpot has finite force, hence finite velocity, hence continuous position: $x_{out}(0^-)=x_{out}(0^+)$).

$$y_c(0^+) = Ae^{-0^+/\tau} + 1 = 0$$
$$= A + 1$$
$$A = -1$$

$$y_c(t) = 1 - e^{-t/\tau}$$

For Knowledge

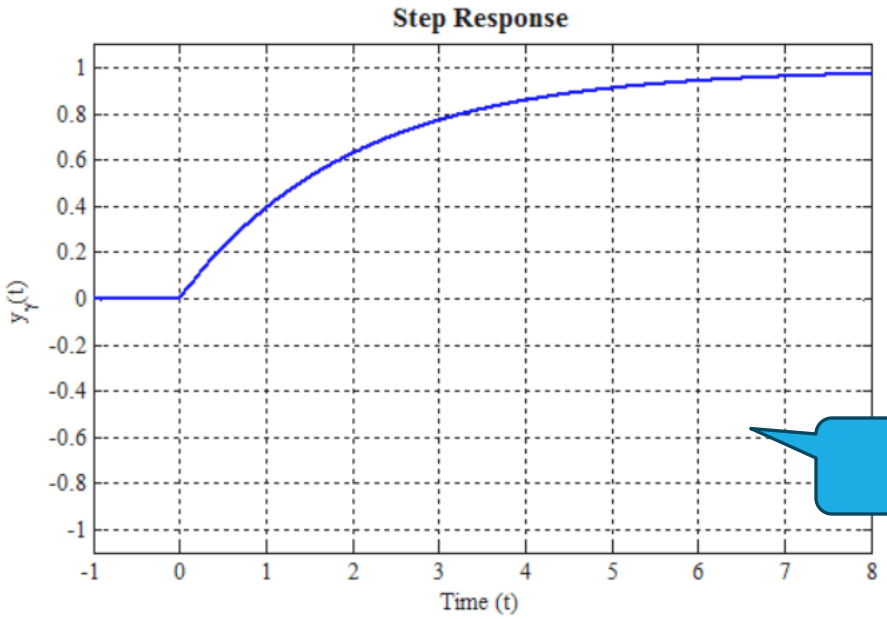
STEP RESPONSE OF A 1ST ORDER SYSTEM

1st Order System

$$x(t) = m\frac{\partial yt)}{\partial t} + ny(t) \qquad H(s) = \frac{a}{s+b}$$

$$y_c(t) = 1 - e^{-t/\tau}$$

The response is shown below for a τ=2 sec.



Overdamped Response

STEP RESPONSE OF A 1ST ORDER SYSTEM

1st Order System

$$x(t) = m\frac{\partial yt)}{\partial t} + ny(t) \qquad H(s) = \frac{a}{s+b}$$

$$v_i(t) = RC\frac{\partial v_o(t)}{\partial t} + v_o(t) \Leftrightarrow \frac{1}{RC}v_i(s)$$

$$\frac{v_o(s)}{v_i(s)} = \frac{\frac{1}{RC}}{s + \frac{1}{RC}}$$

Input is a step function $v_i(s) = \frac{1}{s}$

$$v_o(s) = \frac{\frac{1}{RC}}{s(s + \frac{1}{RC})}$$

$$v_o(t) = 1 - e^{-\frac{1}{RC}t}$$

	$\frac{1}{s^{n+1}}$	
6.	$\frac{1}{s+a}$	e^{-at}
7.	$\frac{1}{(s+a)^n}$	$\frac{1}{(n-1)!}t^{n-1}e^{-at}$
8.	$\frac{b}{s^2+b^2}$	$\sin bt$
9.	$\frac{s}{s^2+b^2}$	$\cos bt$
10.	$\frac{b}{(s+a)^2+b^2}$	$e^{-at}\sin bt$
11.	$\frac{s+a}{(s+a)^2+b^2}$	$e^{-at}\cos bt$
12.	$\frac{a}{s(s+a)}$	$1-e^{-at}$
13.	$\frac{1}{(s+a)(s+b)}$	$\frac{1}{b-a}(e^{-at}-e^{-bt})$

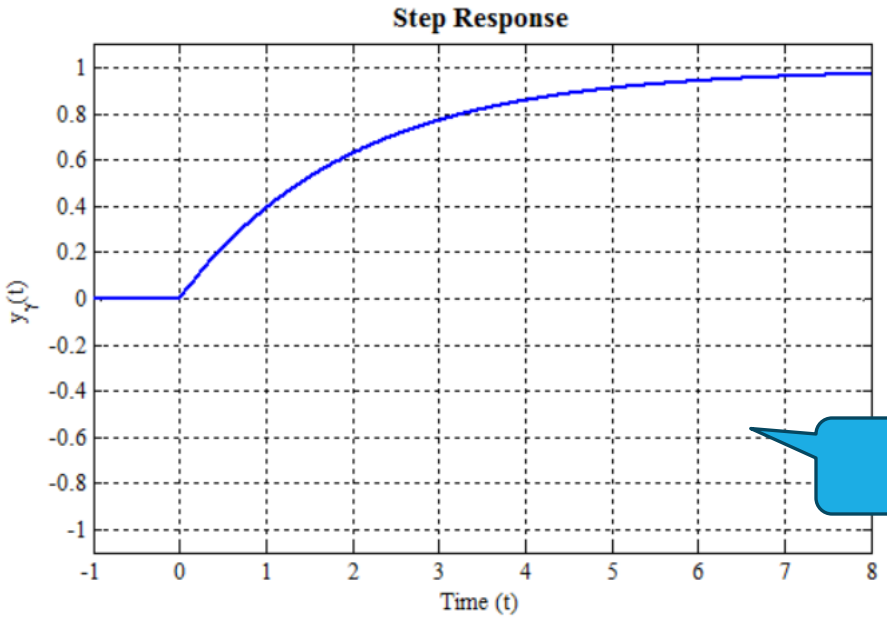
STEP RESPONSE OF A 1ST ORDER SYSTEM

1st Order System

$$x(t) = m\frac{\partial yt)}{\partial t} + ny(t) \qquad H(s) = \frac{a}{s+b}$$

$$y_c(t) = 1 - e^{-t/\tau}$$

The response is shown below for a τ=2 sec.



Overdamped Response

For Knowledge

STEP RESPONSE OF A 2ND ORDER SYSTEM

2nd Order System

$$x(t) = p\frac{\partial^2 y(t)}{\partial t^2} + q\frac{\partial y(t)}{\partial t} + ry(t) \qquad H(s) = \frac{a}{s^2+bs+c}$$

$$\ddot{e}_{out} + \frac{R}{L}\dot{e}_{out} + \frac{1}{LC}e_{out} = \frac{R}{L}\dot{e}_{in}$$

Homogeneous Response of Second Order System

To find the homogeneous solution of a second order system, we proceed as before, i.e., we assume the form of the solution to be

We set the input in th

https://lpsa.swarthmore.edu/Transient/TransInputs/TransStepTime.html

$= 0$

This last equation is called the "characteristic equation" of the system. Since we get a second order polynomial, we get two values for s

$$s_{1,2} = \frac{-\frac{R}{L} \pm \sqrt{\left(\frac{R}{L}\right)^2 - \frac{4}{LC}}}{2}$$

and our homogeneous solution becomes

$$e_{out,h}(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

STEP RESPONSE OF A 2ND ORDER SYSTEM

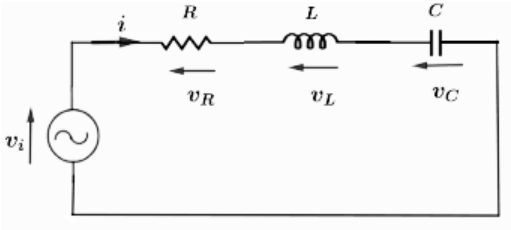
2nd Order System

$$x(t) = p\frac{\partial^2 y(t)}{\partial t^2} + q\frac{\partial y(t)}{\partial t} + ry(t)$$

$$H(s) = \frac{a}{s^2+bs+c}$$

$$\frac{v_o(s)}{v_i(s)} = \frac{\frac{1}{RC}}{s + \frac{1}{RC}}$$

RLC Circuit that we have studied



RLC Network (2nd Order System)

Transfer function

$$\frac{\frac{1}{LC}}{\left(s^2 + \frac{R}{L}s + \frac{1}{LC}\right)}$$

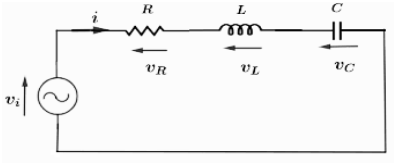
In Laplace Transform, we can learn a lot about the system response from the denominator of the Transfer function

STEP RESPONSE OF A 2ND ORDER SYSTEM

2nd Order System

$$x(t) = p\frac{\partial^2 y(t)}{\partial t^2} + q\frac{\partial y(t)}{\partial t} + ry(t)$$

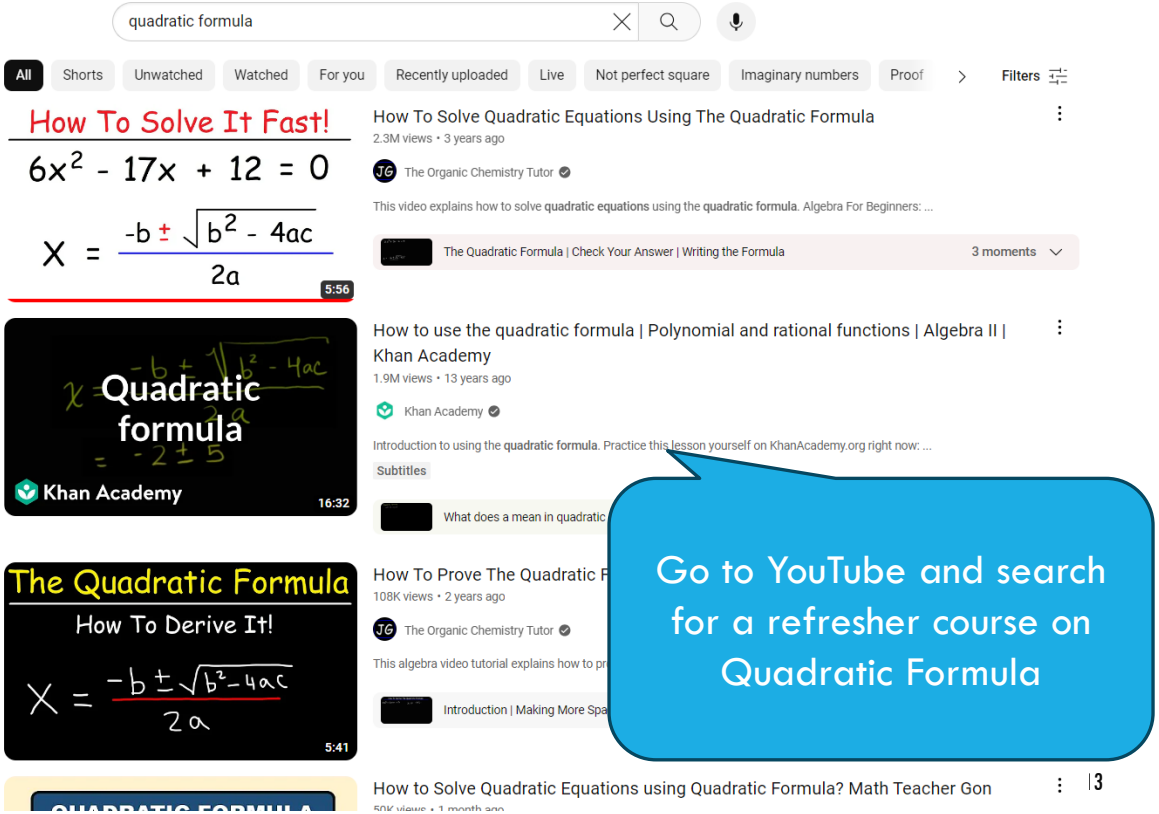
$$H(s) = \frac{a}{s^2+bs+c}$$



RLC Network (2nd Order System)

Transfer function

$$\frac{\frac{1}{LC}}{\left(s^2 + \frac{R}{L}s + \frac{1}{LC}\right)}$$



Go to YouTube and search for a refresher course on Quadratic Formula

STEP RESPONSE OF A 2ND ORDER SYSTEM

2nd Order System

$$x(t) = p\frac{\partial^2 y(t)}{\partial t^2} + q\frac{\partial y(t)}{\partial t} + ry(t)$$

$$H(s) = \frac{a}{s^2+bs+c}$$

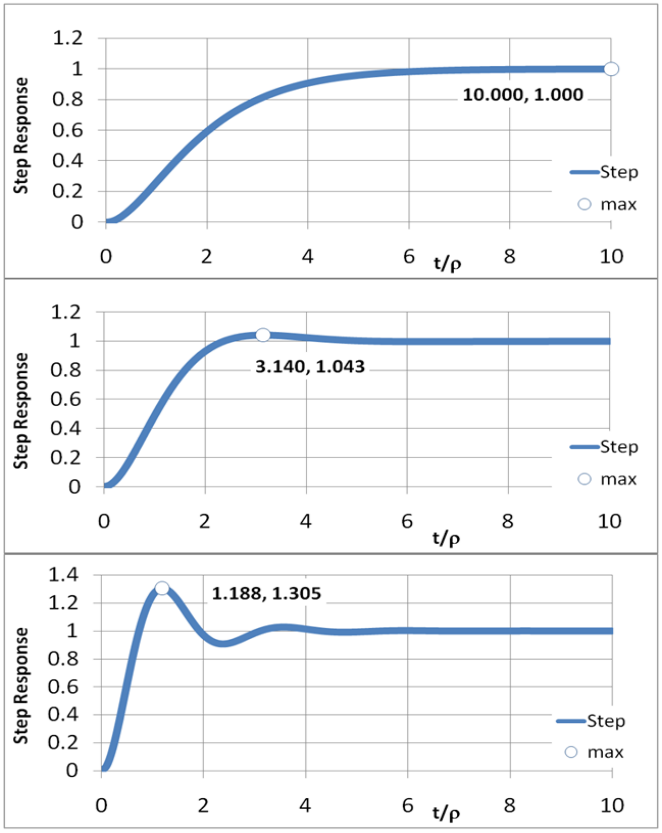
$$\frac{s}{L\left(s^2 + \frac{R}{L}s + \frac{1}{LC}\right)}$$

RLC Network (2nd Order System)

Transfer function

$$\frac{s}{L\left(s^2 + \frac{R}{L}s + \frac{1}{LC}\right)}$$

2nd Order System
(3 unique cases)



STEP RESPONSE OF A 3RD ORDER SYSTEM

3rd Order System

$$x(t) = u\frac{\partial^3 y(t)}{\partial t^3} + v\frac{\partial^2 y(t)}{\partial t^2} + w\frac{\partial y(t)}{\partial t} + zy(t)$$

$$H(s) = \frac{g}{s^3+ds^2+es+f}$$

DC Motor (Speed Control)

3rd Order System
(5 unique cases)

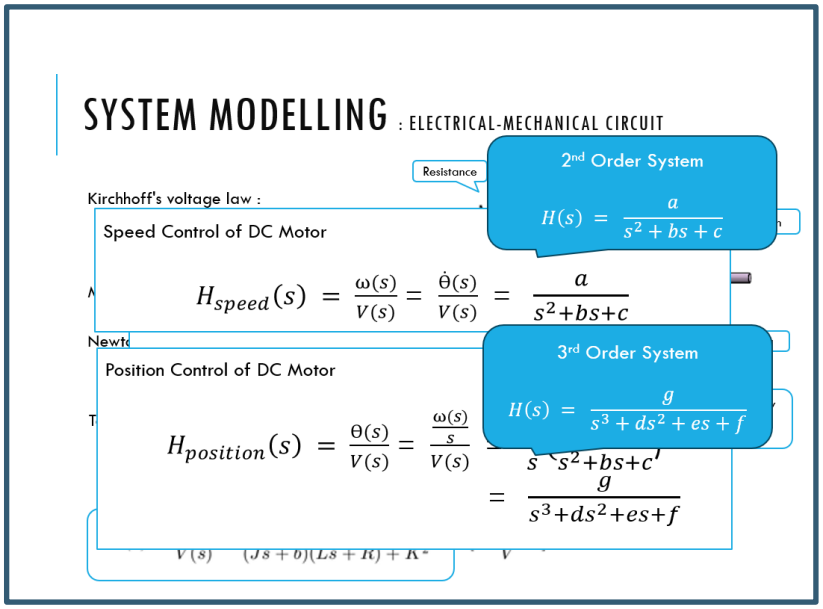
$R) + K^2$

DC Motor (Position Control)

$$\frac{dP_{pos}(t)}{dx} = P_{sp}(t)$$

$$sP_{pos}(s) = P_{sp}(s)$$

$$P_{pos}(s) = \frac{\Theta(s)}{V(s)} = \frac{K}{s((Js+b)(Ls+R)+K^2)}$$



STEP RESPONSE OF A 3RD ORDER SYSTEM

3rd Order System

$$x(t) = u\frac{\partial^3 y(t)}{\partial t^3} + v\frac{\partial^2 y(t)}{\partial t^2} + w\frac{\partial y(t)}{\partial t} + zy(t)$$

$$H(s) = \frac{g}{s^3+ds^2+es+f}$$

DC Motor (Speed Control)

3rd Order System
(5 unique cases)

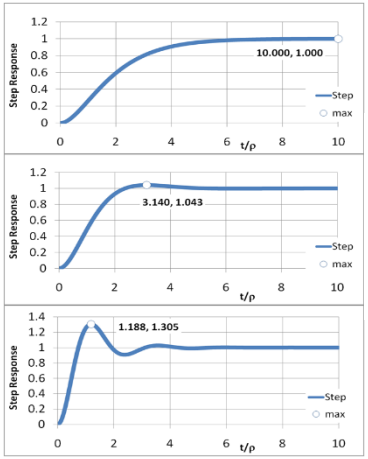
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DC Motor (Position Control)

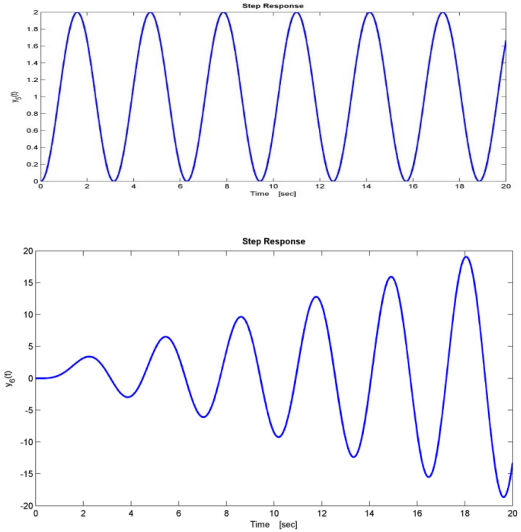
$$\frac{dP_{pos}(t)}{dx} = P_{sp}(t)$$

$$sP_{pos}(s) = P_{sp}(s)$$

$$P_{pos}(s) = \frac{\Theta(s)}{V(s)} = \frac{K}{s((Js+b)(Ls+R)+K^2)}$$



Case 4, 5 :
Marginally Stable, Unstable



STEP RESPONSE (1ST, 2ND, 3RD ORDER SYSTEM)

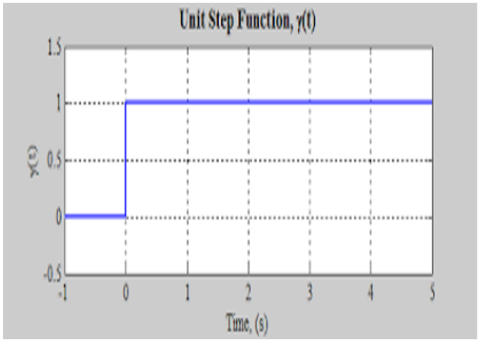
Summary :

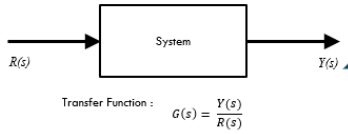
- 1st Order System : The Step Response is always Overdamped. The only difference is the rise time.
- 2nd Order System : The Step Response can be Overdamped, Critically damped or Underdamped (with overshoot). The only difference is the rise time.
- 3rd Order System : The Step Response could include the 3 responses of the 2nd Order System and 2 additional cases of Unstable responses.

Depending on the Parameters of the system. The 1st, 2nd or 3rd Order System will take on 1 of the unique Step Response (of the selection of Step Responses in that particular Order)

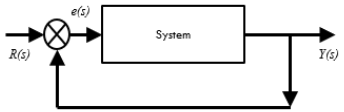
STEP RESPONSE (1ST, 2ND, 3RD ORDER SYSTEM)

STEP RESPONSE





Transfer Function : $G(s) = \frac{Y(s)}{R(s)}$

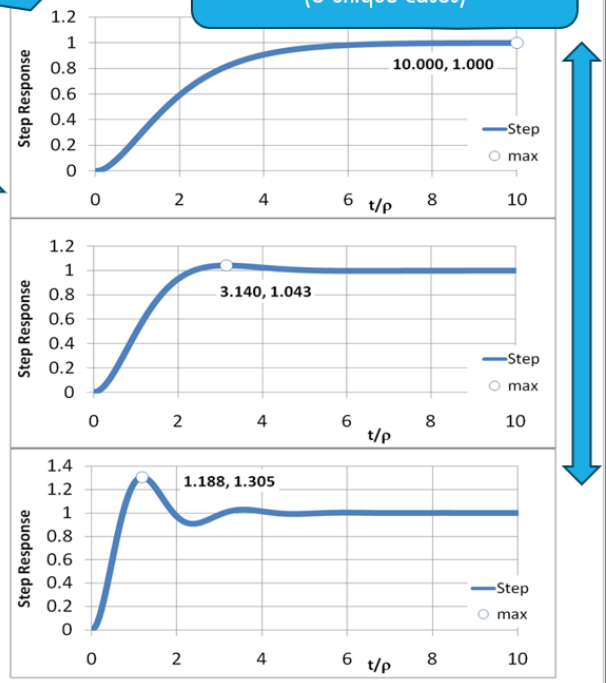


Transfer Function : $T(s) = \frac{G(s)}{1+G(s)}$

1st Order System
(1 unique case)

2nd Order System
(3 unique cases)

3rd Order System
(5 unique cases)



IMPULSE/STEP RESPONSE OF THE SYSTEM

Impulse Response

$$G(s) = \frac{Y(s)}{\delta(s)} = \frac{1}{s^2 + s + 1}$$

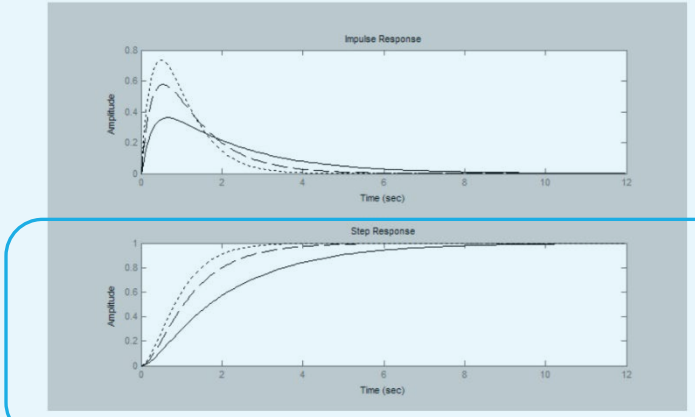
$$Y(s) = \frac{1}{s^2 + s + 1}$$

Unit Step Response

$$G(s) = \frac{Y(s)}{U(s)} = \frac{1}{s^2 + s + 1}$$

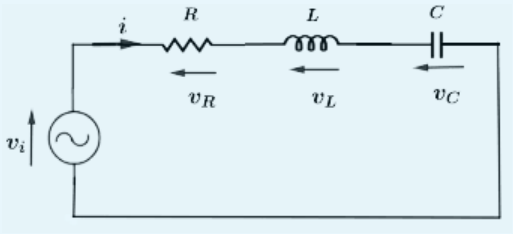
$$Y(s) = \frac{1}{s(s^2 + s + 1)}$$

Overdamped and critically damped system response.



IMPULSE/STEP RESPONSE OF TH

a possible
2nd Order System

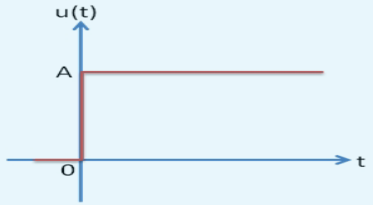


RLC Network (2nd Order System)

Transfer function

$$\frac{\frac{1}{LC}}{\left(s^2 + \frac{R}{L}s + \frac{1}{LC}\right)}$$

$U(t)$ is a step input



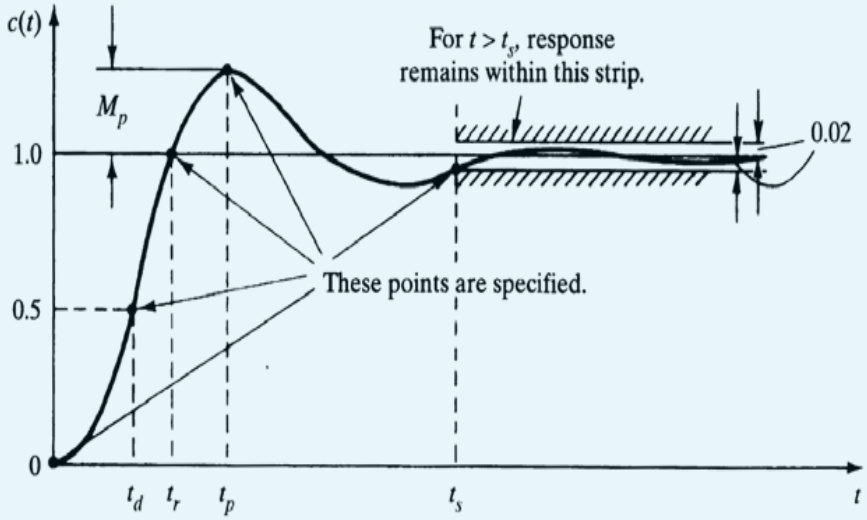
Steady state value, y_∞: the final value of the step response.

Rise time, t_r: The time elapsed up to the instant at which the step response reaches, for the first time, the value $k_r y_\infty$. The constant k_r varies from 0.9 or 1.

Overshoot, M_p: The maximum instantaneous amount by which the step response exceeds its final value.

Settling time, t_s: the time elapsed until the step response enters (without leaving it afterwards) a specified deviation band, $\pm\delta$, around the final value. This deviation δ, is usually say 2% to 5%.

Response : $y(t)$



POLES & ZEROS IN THE LAPLACE TRANSFORM DOMAIN

Transfer Function: Describe the characteristics of the system of interest

$$\frac{\frac{1}{LC}s}{\left(s^2 + \frac{R}{L}s + \frac{1}{LC}\right)} = \frac{\boldsymbol{Zeros}}{\boldsymbol{Poles}}$$

Zeros

Roots of Numerator of the transfer function

$$\frac{1}{LC}s$$

Describe the limits of performance

Poles:

Roots of denominator of the transfer function

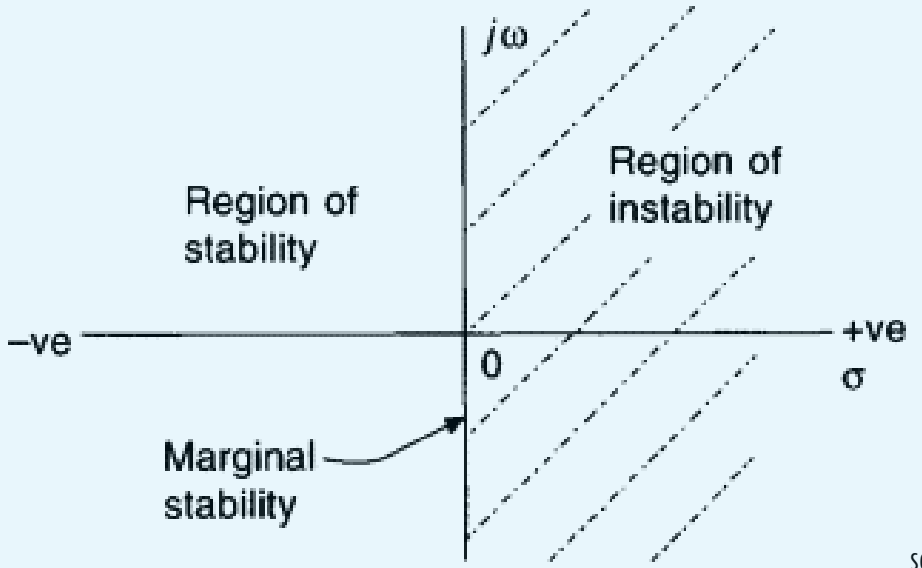
$$L\left(s^2 + \frac{R}{L}s + \frac{1}{LC}\right)$$

Describe the stability of the system

S-PLANE (INTRO TO CONTROL TERM)

S-plane is a two-dimensional space determined by two orthogonal axes, the real number axis and the imaginary number axis. A point in the S-plane represents a complex number.

A point on the real number axis represents only real value. A point on the imaginary axis represents only imaginary value



IMPULSE/STEP RESPONSE (2ND ORDER SYSTEM)

damping ratio, ω_n (ω_o) = natural undamped frequency (Intro to control term)

$$Y_\gamma(s) = \frac{1}{s}H(s) = \frac{1}{s}K\frac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2}$$

$as^2 + bs + c$

ζ = damping ratio ω_n (ω_o) = natural undamped frequency

To solve for $y_\gamma(t)$, we need to find the poles (which are the roots of the denominator of the transfer function

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$s = \frac{-2\zeta\omega_0 \pm \sqrt{(2\zeta\omega_0)^2 - 4\omega_0^2}}{2} = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$$

IMPULSE/STEP RESPONSE (2ND ORDER SYSTEM)

damping ratio, ω_n (ω_o) = natural undamped frequency (Intro to control term)

The value of ζ (or roots of characteristic equation) determine five unique cases of interest that are given special names

$$s = \frac{-2\zeta\omega_0 \pm \sqrt{(2\zeta\omega_0)^2 - 4\omega_0^2}}{2} = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$$

3+2 unique cases (2nd order system)

Name	Value of ζ	Roots of s	Characteristics of "s"
Overdamped	$\zeta>1$	$s = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$	Two real and negative roots
Critically Damped	$\zeta=1$	$s = -\omega_0$	A single (repeated) negative root
Underdamped	$0<\zeta<1$	$s = -\zeta\omega_0 \pm j\omega_0\sqrt{1 - \zeta^2}$	Complex conjugate ($j = \sqrt{-1}$);
Undamped	$\zeta=0$	$s = \pm j\omega_0$	Pure imaginary (no real part)
Exponential Growth	$\zeta<0$	$s = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$	Roots may be complex or real, but the real part of s is always positive

LET ANALYSIS (SECOND-ORDER SYSTEM)

If 2nd Order System

$$\frac{\partial^2 y(t)}{\partial x(t)^2} + m\frac{\partial y(t)}{\partial x(t)} + ny(t) \quad \leftrightarrow \text{differential equation}$$

Transfer Function $G(s) = \frac{Y(s)}{X(s)} = \frac{(s \pm p)}{(s^2+qs+r)}$ ↔ algebraic equation

$$G(s) = \frac{Y(s)}{X(s)} = \frac{(s \pm p)}{(s^2+qs+r)} = \frac{zeros}{poles}$$

Characteristic Equation (or System Characteristic Equation is very Important)

LET ANALYSIS (SECOND-ORDER SYSTEM) CASE-01

$$G(s) = \frac{Y(s)}{X(s)} = \frac{(s \pm p)}{(s^2 + qs + r)} = \frac{Zeros}{Poles}$$

System Characteristic Equation

$$s^2 + 5s + 6$$

$$(s + 3)(s + 2)$$

Two distinct Roots
(or 2 distinct Poles)

LET ANALYSIS (SECOND-ORDER SYSTEM) CASE-01

Case 1: The overdamped case (ζ>1)

In the overdamped case we have two real poles of the transfer function (or zeros of the characteristic equation) at

$$s = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$$

For convenience, we will refer to these as α_1 and α_2 (these are shown in a pole-zero diagram at the right)

$$\alpha_1 = \zeta\omega_0 + \omega_0\sqrt{\zeta^2 - 1} = \omega_0\left(\zeta + \sqrt{\zeta^2 - 1}\right)$$

$$\alpha_2 = \zeta\omega_0 - \omega_0\sqrt{\zeta^2 - 1} = \omega_0\left(\zeta - \sqrt{\zeta^2 - 1}\right)$$

and note that

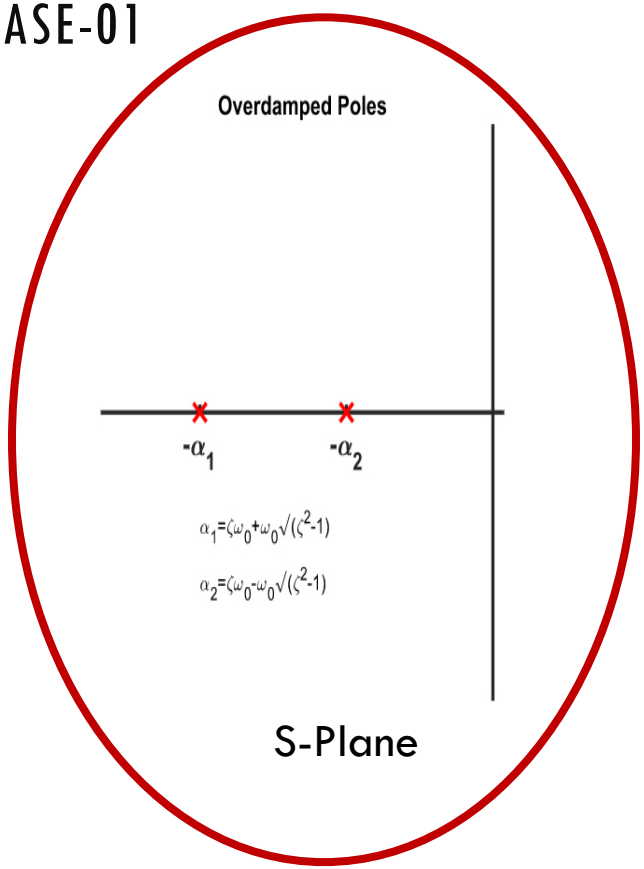
$$\alpha_1 \cdot \alpha_2 = \omega_0\left(\zeta + \sqrt{\zeta^2 - 1}\right) \cdot \omega_0\left(\zeta - \sqrt{\zeta^2 - 1}\right) = \omega_0^2\left(\zeta^2 + -\left(\zeta^2 - 1\right) + \zeta\sqrt{\zeta^2 - 1} - \zeta\sqrt{\zeta^2 - 1}\right) = \omega_0^2$$

The transfer function may now be written as

$$H(s) = K\frac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2} = K\frac{\alpha_1 \cdot \alpha_2}{(s + \alpha_1)(s + \alpha_2)}$$

and the unit step response as

$$Y_\gamma(s) = \frac{1}{s}H(s) = \frac{1}{s}K\frac{\alpha_1 \cdot \alpha_2}{(s + \alpha_1)(s + \alpha_2)}$$



LET ANALYSIS (SECOND-ORDER SYSTEM) CASE-02

$$G(s) = \frac{Y(s)}{X(s)} = \frac{(s \pm p)}{(s^2 + qs + r)} = \frac{Zeros}{Poles}$$

System Characteristic Equation

$$s^2 + 6s + 9$$

$$(s + 3)(s + 3)$$

Repeated Roots
(or Repeated Poles)

LET ANALYSIS (SECOND-ORDER SYSTEM) CASE-02

Case 2: The critically damped case (ζ=1)

To find the response of the critically damped case we proceed as with the overdamped case. For ζ=1 the roots of the denominator of the transfer function are both at s=-ω_0 (an alternative notations has poles at s=-α) so the transfer function can be written as

$$H(s) = K\frac{\omega_0^2}{s^2 + 2\omega_0 s + \omega_0^2} = K\frac{\omega_0^2}{(s+\omega_0)^2} = K\frac{\alpha^2}{(s+\alpha)^2}$$

This has a repeated pole at s=-ω_0 (or equivalently, s=-α as shown on the pole-zero diagram.

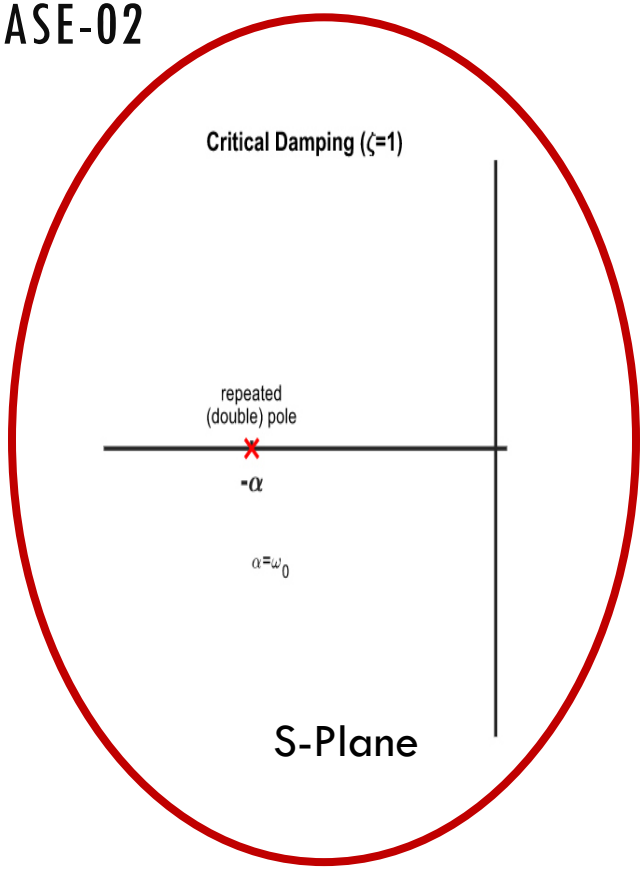
$$Y_\gamma(s) = \frac{1}{s}H(s) = \frac{1}{s}K\frac{\omega_0^2}{(s+\omega_0)^2} = \frac{1}{s}K\frac{\alpha^2}{(s+\alpha)^2}$$

This is the "asymptotic critically damped" form in the Laplace transform table, so

$$\begin{aligned} y_\gamma(t) &= K\left(1 - e^{-\omega_0 t} - \omega_0 t e^{-\omega_0 t}\right) \\ &= K\left(1 - e^{-\alpha t} - \omega_0 t e^{-\alpha t}\right) \end{aligned}$$

We can note several characteristics of this response:

1. The final value as t→∞ is K, the system gain. This is also H(0).
2. The inital value as t→0^+ is 0. This is also H(∞).
3. Wherever ω_0 occurs, it is multiplied by t. That means if we double ω_0 we double the speed of the system, but don't change the shape of the response.
4. As the poles move away from the origin (α increases) the speed of the system gets faster.



LET ANALYSIS (SECOND-ORDER SYSTEM) CASE-03

$$G(s) = \frac{Y(s)}{X(s)} = \frac{(s \pm p)}{(s^2 + qs + r)} = \frac{Zeros}{Poles}$$

System Characteristic Equation

$$s^2 + 25s + 344$$

$$(s + 12.5 + j13.7)(s + 12.5 - j13.7)$$

2 Complex Roots
(or 2 Complex Poles)

LET ANALYSIS (SECOND-ORDER SYSTEM) CASE-03

Case 3: The underdamped case (ζ<1)

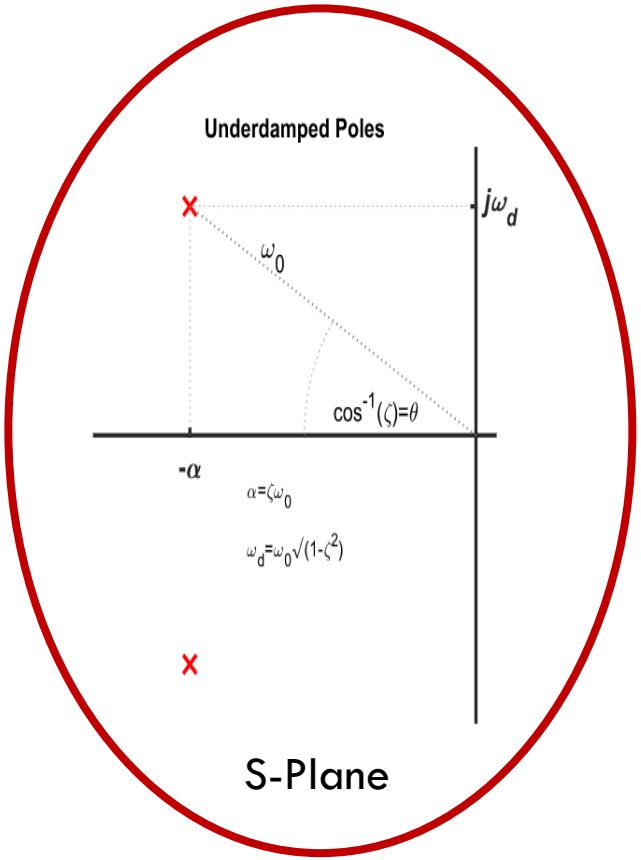
For the underdamped case we use the transfer function to find the step response in the Laplace domain. The pole locations can either be written in terms of ζ and ω_0, or in terms of their real and imaginary parts (α and ω_d),

$$H(s) = K\frac{\omega_0^2}{s^2 + 2\omega_0 s + \omega_0^2} = K\frac{\alpha^2 + \omega_d^2}{(s+\alpha)^2 + \omega_d^2}$$

These pole locations are shown on the diagram at right. Note that -α is the real part of the pole location, and $\pm j\omega_d$ is the imaginary part. Or we can say that ω_0 is the distance from the origin, and the angle the pole makes with the horizontal axis is determined by ζ (θ=acos(ζ)); We can relate these two representations by:

$$\omega_d = \omega_0\sqrt{1-\zeta^2}$$

$$\alpha = \zeta\omega_0$$



LET ANALYSIS (SECOND-ORDER SYSTEM) CASE-03

The step response is

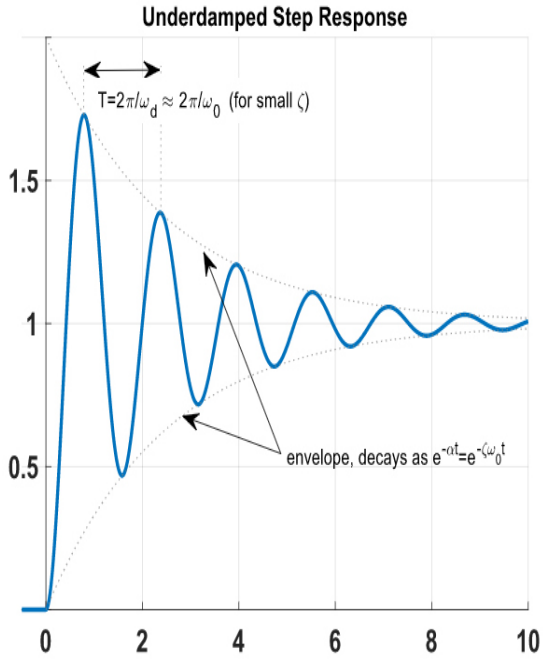
$$Y_\gamma(s) = \frac{1}{s}H(s) = \frac{1}{s}K\frac{\omega_0^2}{s^2+2\omega_0 s+\omega_0^2} = \frac{1}{s}K\frac{\alpha^2+\omega_d^2}{(s+\alpha)^2+\omega_d^2}$$

We find this form in the Laplace transform table ("Prototype 2nd order lopass step response"), so

$$y_\gamma(t) = K\left(1-\frac{1}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_0 t}\sin\left(\omega_0\sqrt{1-\zeta^2}t+\theta\right)\right)$$

$$= K\left(1-\frac{\sqrt{\alpha^2+\omega_d^2}}{\omega_d}e^{-\alpha t}\sin(\omega_d t+\theta)\right)$$

$$\theta = \operatorname{acos}(\zeta) = \operatorname{atan}\left(\frac{\omega_d}{\alpha}\right)$$



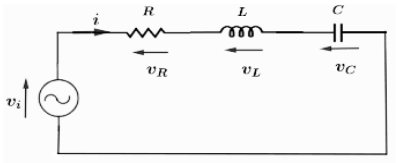
There is a lot of information in this expression. Several important characteristics of the equation include:

1. The final value as $t\to\infty$ is K, the system gain. This is also H(0).
2. The inital value as $t\to 0^+$ is 0. This is also $H(\infty)$.
3. Wherever ω_0 occurs, it is multiplied by t. That means if we double ω_0 we double the speed of the system, but don't change the shape of the response. As ω_0 increases the system becomes faster, and the poles move away from the origin.
4. The "decay" ($e^{-\zeta\omega 0t}$) has a decay coefficient of $\alpha=\zeta\omega_0$ (or a time time constant $\tau=1/(\zeta\omega_0)$.
5. The frequency of oscillation is called the damped frequency, ω_d, where $\omega_d = \omega_0\sqrt{1-\zeta^2}$. *Note: for small ζ, $\omega_d\approx\omega_0$. (For example, if $\zeta=0.2$, $\omega_d=0.98\omega_0$; if $\zeta=0.4$, $\omega_d=0.92\omega_0$.*

LET ANALYSIS (SECOND-ORDER SYSTEM) CASE-04

Mathematically Possible Case

$$G(s) = \frac{Y(s)}{X(s)} = \frac{(s \pm p)}{(s^2 + qs + r)} = \frac{Zeros}{Poles}$$



RLC Network (2nd Order System)

Transfer function

$$\frac{s}{L\left(s^2 + \frac{R}{L}s + \frac{1}{LC}\right)}$$

System Characteristic Equation

$$s^2 + 0s + 100$$

$$(s + j10)(s - j10)$$

complex roots

LET ANALYSIS (SECOND-ORDER SYSTEM) CASE-04

Case 4: The undamped case (ζ=0)

When the damping coefficient is zero the system is said to be undamped. The roots of the denominator of the transfer function are at $s=\pm j\omega_0$ so the transfer function is

$$H(s) = K\frac{\omega_0^2}{s^2+\omega_0^2} = K\frac{\omega_0^2}{(s+j\omega_0)(s-j\omega_0)}$$

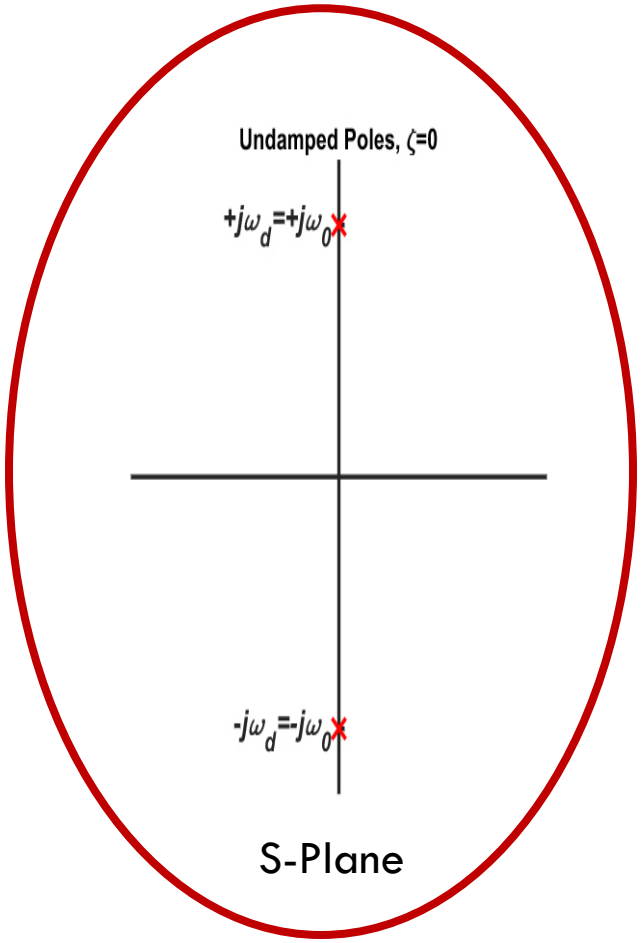
which give a step response

$$Y_\gamma(s) = \frac{1}{s}H(s) = \frac{1}{s}K\frac{\omega_0^2}{(s+\omega_0)^2}$$

This is a special case of the "Prototype 2nd order lopass step response" form in the Laplace transform table with ζ=0. So we get:

$$y_\gamma(t) = K\left(1-\sin(\omega_0 t+\pi)\right) = K\left(1-\cos(\omega_0 t)\right)$$

As expected from the name, the undamped system (ζ=0) has no damping and oscillates forever as the graph below shows.



LET ANALYSIS (SECOND-ORDER SYSTEM) CASE-05

Mathematically Possible Case

$$G(s) = \frac{Y(s)}{X(s)} = \frac{(s \pm p)}{(s^2 + qs + r)} = \frac{Zeros}{Poles}$$

System Characteristic Equation

$$s^2 - 2s + 5$$

$$(s - 1 + j2)(s - 1 - j2)$$

real part of roots on right-hand side of S-plane

2 Complex Roots (or 2 Complex Poles)

LET ANALYSIS (SECOND-ORDER SYSTEM) CASE-05

Case 5: Exponential growth (ζ<0)

If we consider the case where ζ is negative, we can write the transfer function in terms of the two roots of the denominator of the transfer function

$$H(s) = K\frac{\omega_0^2}{s^2 + 2\zeta\omega_0 s + \omega_0^2} = K\frac{\alpha_1 \cdot \alpha_2}{(s+\alpha_1)(s+\alpha_2)}$$

$$\alpha_1 = -\zeta\omega_0 + \omega_0\sqrt{\zeta^2 - 1} = \omega_0\left(-\zeta + \sqrt{\zeta^2 - 1}\right)$$

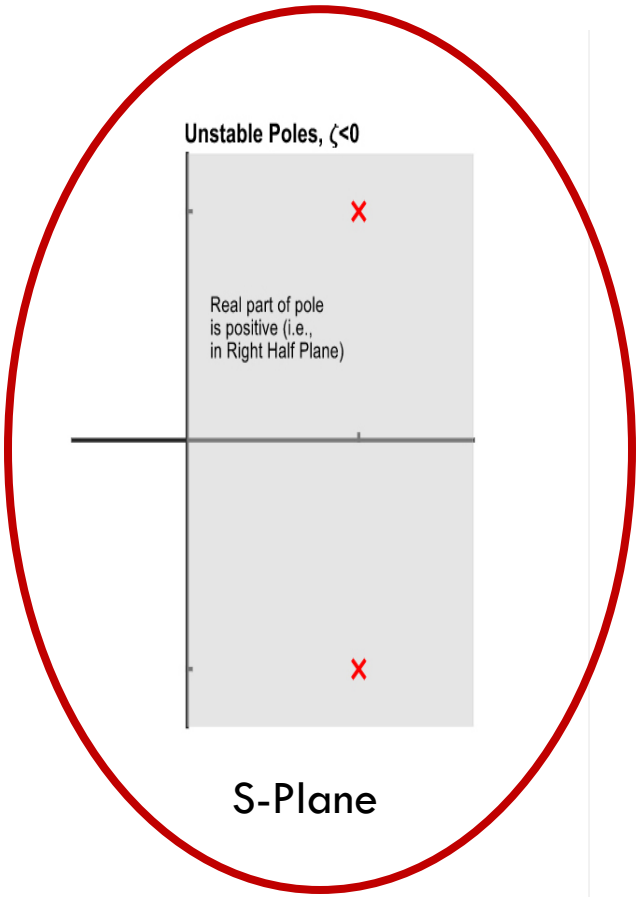
$$\alpha_2 = -\zeta\omega_0 - \omega_0\sqrt{\zeta^2 - 1} = \omega_0\left(-\zeta - \sqrt{\zeta^2 - 1}\right)$$

We determine the unit step response by multiplying H(s) by 1/s (a unit step input), and performing a partial fraction expansion (assuming, for now, that α_1 and α_2 are not equal):

$$Y_\gamma(s) = \frac{1}{s}H(s) = \frac{1}{s}K\frac{\alpha_1 \cdot \alpha_2}{(s+\alpha_1)(s+\alpha_2)}$$

$$= \frac{A_1}{s} + \frac{A_2}{s+\alpha_1} + \frac{A_3}{s+\alpha_2}$$

$$y_\gamma(t) = A_1 + A_2 e^{-\alpha_1 t} + A_3 e^{-\alpha_2 t}$$



Note we have not calculated the values of A_1, A_2, and A_3, but their precise values are unimportant to the argument being made; i.e., that the solution grows exponentially.

Since the real parts of α_1 and α_2 are negative, this solution grows exponentially as t increases. If α_1 and α_2 are complex (as shown in the pole-zero diagram), the solution oscillates as it grows. For the types of systems we will discuss, this behavior is rare. However, they are important in the topic of "Control Theory.".

IMPULSE/STEP RESPONSE (2ND ORDER SYSTEM)

A family of unit-step response curves $c(t)$ with various values of z is shown in Figure1 , where the abscissa is the dimensionless variable $\omega_n t$.

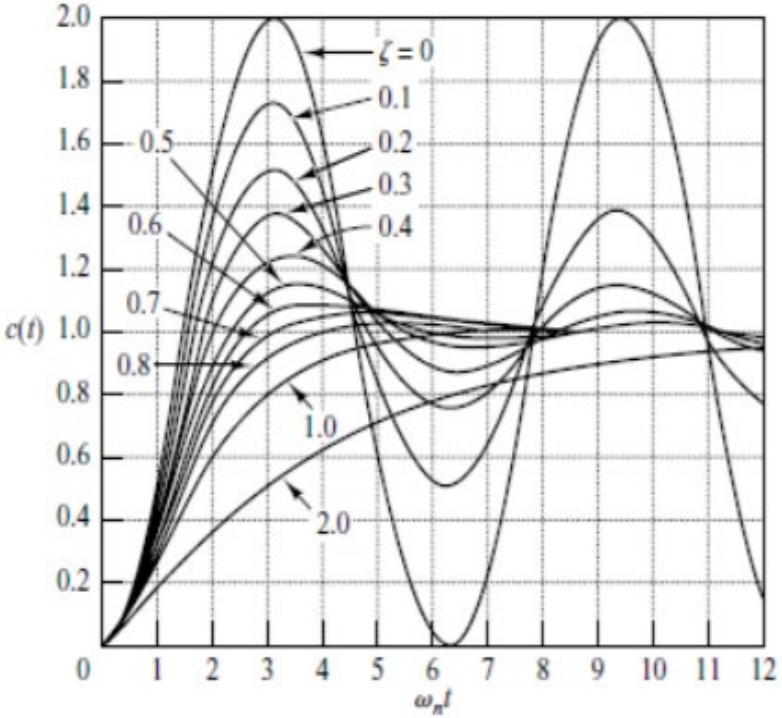
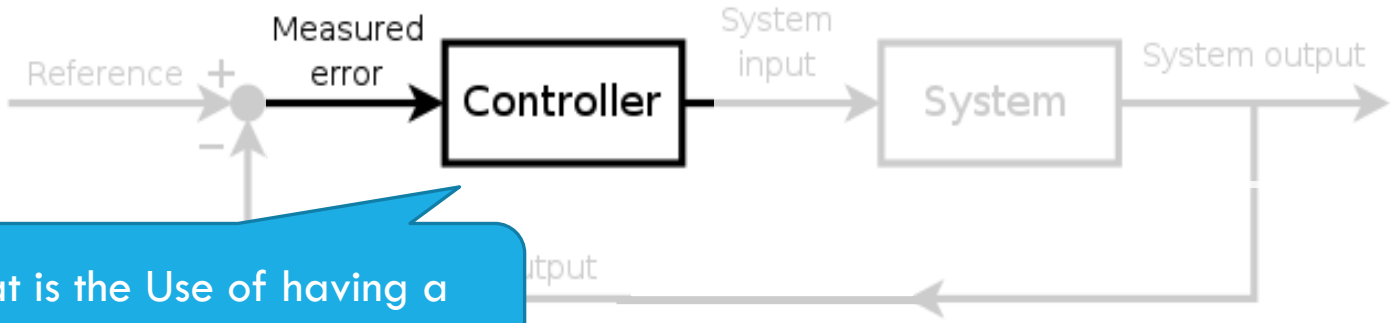


Figure 1: Unit step response curves of the system

SYSTEM BLOCK DIAGRAM (OPEN/CLOSED LOOP)



What is the Use of having a Controller ?

Overall Transfer Function

(if we consider a simple Proportional Controller, K)

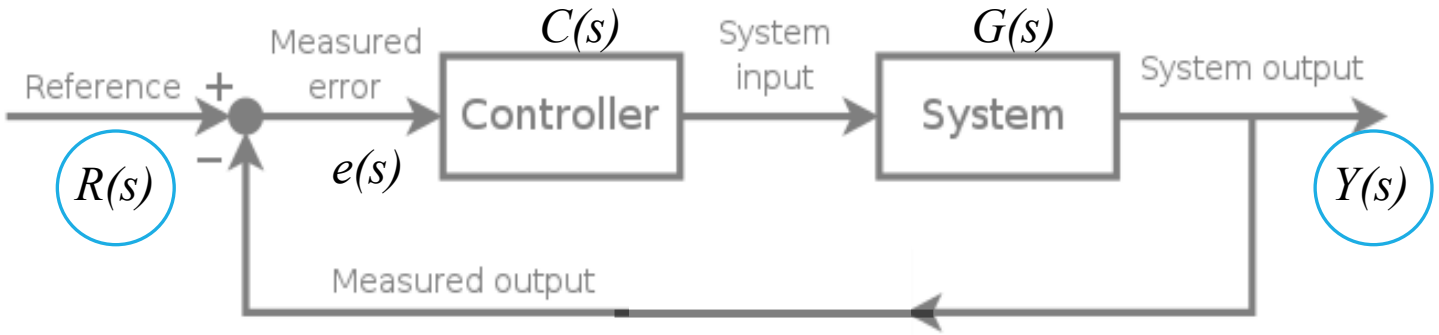
$$\frac{Y(s)}{R(s)} = \frac{KG(s)}{1+KG(s)}$$

↔ algebraic equation

Basic Controller :

Proportional Controller (***K***)

SYSTEM BLOCK DIAGRAM (CLOSED LOOP)



$$Y(s) = C(s)G(s)e(s)$$
$$= KG(s)e(s)$$

$$R(s) - Y(s) = e(s)$$

$$Y(s) = KG(s)\,[R(s) - Y(s)]$$
$$Y(s)[1 + KG(s)] = KG(s)R(s)$$

Overall Transfer Function *(if we consider a simple Proportional Controller, K)*

$$\frac{Y(s)}{R(s)} = \frac{KG(s)}{1+KG(s)}$$

↔ algebraic equation

Derivation of the
System Transfer Function

RESPONSE DON'T MEET SPECIFICATIONS?



Open Loop System

Transfer Function :

$$G(s) = \frac{Y(s)}{R(s)} = \frac{1}{(s+1)(s+9)}$$

Name	Value of ζ	Roots of s	Characteristics of "s"
Overdamped	ζ>1	$s = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$	Two real and negative roots
Critically Damped	ζ=1	$s = -\omega_0$	A single (repeated) negative root
Underdamped	0<ζ<1	$s = -\zeta\omega_0 \pm j\omega_0\sqrt{1 - \zeta^2}$	Complex conjugate ($j = \sqrt{-1}$);
Undamped	ζ=0	$s = \pm j\omega_0$	Pure imaginary (no real part)
Exponential Growth	ζ<0	$s = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$	Roots may be complex or real, but the real part of s is always positive

RESPONSE DON'T MEET SPECIFICATIONS?

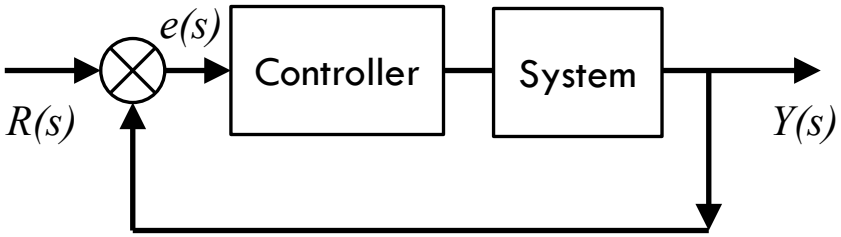


Open Loop System

Transfer Function :

$$G(s) = \frac{Y(s)}{R(s)} = \frac{1}{(s+1)(s+9)}$$

Adding a Controller



Closed Loop Unity Gain Negative Feedback System

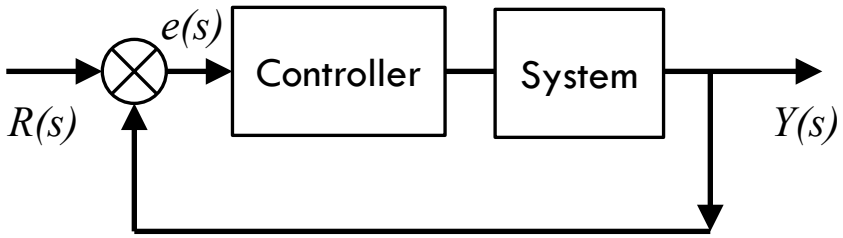
Transfer Function :

$$T(s) = \frac{C(s)G(s)}{1 + C(s)G(s)}$$

RESPONSE DON'T MEET SPECIFICATIONS?

Consider the Example of using a Simple Proportional Controller

Simple Proportional Controller, ***K***



Transfer Function :

$$T(s) = \frac{\boldsymbol{K}G(s)}{1 + \boldsymbol{K}G(s)}$$

$$T(s) = \frac{\boldsymbol{K}\dfrac{1}{(s+1)(s+9)}}{1 + \boldsymbol{K}\dfrac{1}{(s+1)(s+9)}}$$

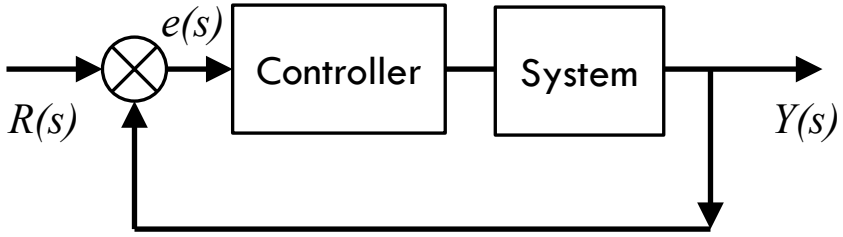
$$T(s) = \frac{\boldsymbol{K}}{(s+1)(s+9) + \boldsymbol{K}}$$

$$T(s) = \frac{\boldsymbol{K}}{s^2 + 10s + 9 + \boldsymbol{K}}$$

RESPONSE DON'T MEET SPECIFICATIONS?

Simple Proportional Controller
$K = 16$

Simple Proportional Controller, K



Transfer Function :

$$T(s) = \frac{K}{s^2 + 10s + 9 + K}$$

Consider the case when the gain $K = 16$

$$T(s) = \frac{16}{s^2 + 10s + 9 + 16} = \frac{16}{(s+5)^2}$$

RESPONSE DON'T MEET SPECIFICATIONS?

Simple Proportional Controller, ***K***

Name	Value of ζ	Roots of s	Characteristics of "s"
Overdamped	ζ>1	$s = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$	Two real and negative roots
Critically Damped	ζ=1	$s = -\omega_0$	A single (repeated) negative root
Underdamped	0<ζ<1	$s = -\zeta\omega_0 \pm j\omega_0\sqrt{1 - \zeta^2}$	Complex conjugate ($j = \sqrt{-1}$);
Undamped	ζ=0	$s = \pm j\omega_0$	Pure imaginary (no real part)
Exponential Growth	ζ<0	$s = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1}$	Roots may be complex or real, but the real part of s is always positive

Consider the case when the gain ***K*** = 16

$$T(s) = \frac{16}{s^2 + 10s + 9 + 16} = \frac{16}{(s+5)^2}$$

RESPONSE DON'T MEET SPECIFICATIONS? (SUMMARY)

Open-Loop System :

$$G(s) = \frac{Y(s)}{R(s)} = \frac{1}{(s+1)(s+9)}$$

Step Response is Case 01 :

Sluggish Overdamped System

Closed-Loop System (with a Proportional Controller, ***K***) :

$$T(s) = \frac{\boldsymbol{K}}{s^2 + 10s + 9 + \boldsymbol{K}}$$

Consider the case when the gain ***K*** = 16

$$T(s) = \frac{16}{s^2 + 10s + 9 + 16} = \frac{16}{(s+5)^2}$$

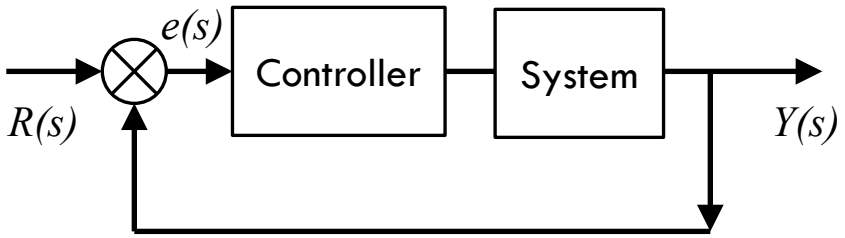
Step Response is Case 02 :

Critically Damped System

RESPONSE DON'T MEET SPECIFICATIONS?

Consider the Example of using a PID (proportional integral differential) Controller

PID Controller $\frac{K(s+1)}{(s+20)}$,



Transfer Function :

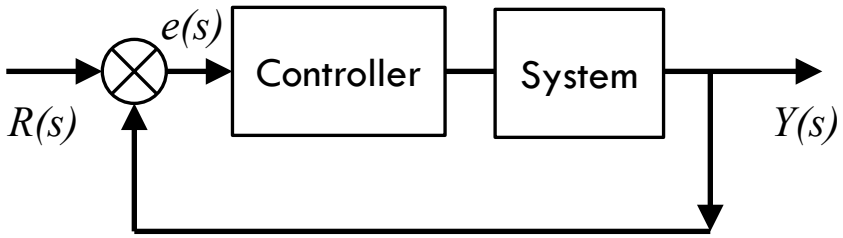
$$T(s) = \frac{C(s)G(s)}{1 + C(s)G(s)}$$

$$T(s) = \frac{\frac{K(s+1)}{(s+20)}\frac{1}{(s+1)(s+9)}}{1 + \left[\frac{K(s+1)}{(s+20)}\frac{1}{(s+1)(s+9)}\right]}$$

$$T(s) = \frac{K}{(s+9)(s+20)+K}$$

RESPONSE DON'T MEET SPECIFICATIONS?

PID Controller $\frac{\boldsymbol{K}(s+1)}{(s+20)}$,



Transfer Function :

$$T(s) = \frac{C(s)G(s)}{1 + C(s)G(s)}$$

$$T(s) = \frac{\frac{\boldsymbol{K}(s+1)}{(s+20)}\frac{1}{(s+1)(s+9)}}{1 + [\frac{\boldsymbol{K}(s+1)}{(s+20)}\frac{1}{(s+1)(s+9)}]}$$

$$T(s) = \frac{\boldsymbol{K}}{(s+9)(s+20) + \boldsymbol{K}}$$

Damping ratio ζ closer to 1

If $\boldsymbol{K}$ is chosen to be very close to zero

CONTROLLER DESIGN (ANALOGUE)

More Formal Controller design is beyond the scope of this SCSE subject. If there is time, Analogue Controller Design Techniques can look at :

- Bode Plot Design
- Root Locus Plot Design

Stability Design & Testing can include those :

- Routh Hurwitz Stability Criterion
- Jury Stability Criterion

EXTRA

TIME DOMAIN SPECIFICATIONS DESIGN

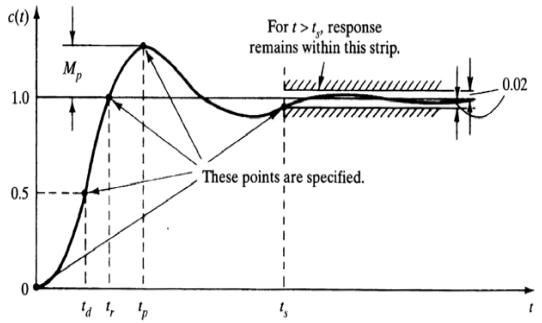
Consider a second-order system to understand the characteristics

$$G(s)=\frac{q(s)}{u(s)}=\frac{w_n^2}{s^2+2\varsigma w_n s+w_n^2}$$

ζ - damping; w_n - natural frequency.

Percentage Maximum Overshoot

$$PM_p=\frac{M_p}{y_\infty}\times 100=100\times\exp\left(\frac{-\varsigma\pi}{\sqrt{1-\varsigma^2}}\right)$$



Rise Time

$$T_r=\frac{\pi-\theta}{w_n\sqrt{1-\varsigma^2}};\ \theta=\cos^{-1}\varsigma$$

Settling Time

$$T_s=\frac{4}{\varsigma w_n}$$

Peak Time

$$T_p=\frac{\pi}{w_n\sqrt{1-\varsigma^2}}$$

TIME DOMAIN SPECIFICATIONS DESIGN

Determine the values of T_p, T_r, T_s, PM_p for the following system.

$$G(s)=\frac{q(s)}{u(s)}=\frac{1}{s^2+s+1}$$

Roots

S = −0.5+j0.866
S = −0.5−j0.866

Solution:

$$s^2+2\varsigma w_n s+w_n^2=s^2+s+1$$

$$\text{Hence, } w_n=1;\ 2\varsigma w_n=1\rightarrow\varsigma=0.5$$

- Maximum peak overshoot

$$PM_p=100\times\exp\left(\frac{-\varsigma\pi}{\sqrt{1-\varsigma^2}}\right)$$

$$=100\times\exp\left(\frac{\pi 0.5}{\sqrt{1-0.5^2}}\right)=16.3\%$$

Peak time

$$T_p=\frac{\pi}{w_n\sqrt{1-\varsigma^2}}=\frac{\pi}{1\sqrt{1-0.5^2}}$$

$$=3.63s$$

TIME DOMAIN SPECIFICATIONS DESIGN

Rise time

$$T_r = \frac{\pi - \theta}{w_n \sqrt{1-\varsigma^2}}; \; \theta = \cos^{-1} \varsigma$$

$$\theta = \cos^{-1} 0.5 = 1.05 \text{ radians}$$

$$T_r = \frac{\pi - 1.05}{1\sqrt{1-0.5^2}} = 2.41s$$

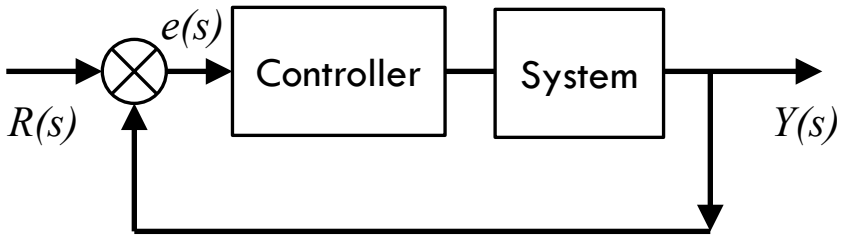
Settling time

$$T_s = \frac{4}{\varsigma w_n} = \frac{4}{0.5 \times 1} = 8s$$

RESPONSE DON'T MEET SPECIFICATIONS?

Reminder

Simple Proportional Controller, ***K***



Transfer Function :

$$T(s) = \frac{\boldsymbol{K}G(s)}{1 + \boldsymbol{K}G(s)}$$

$$T(s) = \frac{\boldsymbol{K}\dfrac{1}{(s+1)(s+9)}}{1 + \boldsymbol{K}\dfrac{1}{(s+1)(s+9)}}$$

$$T(s) = \frac{\boldsymbol{K}}{(s+1)(s+9) + \boldsymbol{K}}$$

$$T(s) = \frac{\boldsymbol{K}}{s^2 + 10s + 9 + \boldsymbol{K}}$$